

3rd-Year Group Design Project

Project Statement

Project Name: Helicopter Emergency Rescue (HER25)

Academic Advisors: Drs. Armanini, Fasel, Ribera Vicent (coordinators), Shamsuddin, Seifikar and Steiros

Duration: 12/05/2025 – 19/06/2025

Project Overview

Project Description

Helicopter Emergency Medical Services (HEMS) are crucial for providing rapid medical response, especially in situations where ground transport is too slow or inaccessible. HEMS plays a vital role in saving lives by quickly reaching accident scenes, transporting critically ill patients, and ensuring timely medical interventions. It is particularly essential in rural, mountainous, or congested urban areas where rapid evacuation can make the difference between life and death.

Current HEMS aircraft are existing helicopters retrofitted for medical missions. While highly capable, they have design drawbacks that limit their effectiveness, such as limited cabin space, noise and vibration issues, performance limitations (in hot-and-high conditions or adverse weather), range constraints, operating and maintenance costs, safety concerns and environmental constraints (e.g. noise levels).

The aim of this project is to design the perfect rotorcraft for HEMS operations, providing optimised cabin layout for medical operations, enhance safety features, performance improvements, noise and vibration reduction, cost and maintenance efficiency and all-weather operations.

Vehicle Main Specifications

The design must meet all relevant airworthiness and safety requirements.

Aircraft Requirements:

- Speed: quick start up time (call to lift), fast cruise (as fast and efficient as possible) and quick turnaround (refuel speed)
- Size – as small a footprint as possible to land as close as needed to patient, but with sufficient interior space for a pilot, a doctor, a paramedic, the patient and a companion (i.e. for minors) or a trainee, and all the medical equipment, with easy access for staff and stretchers, including side or rear access.

A spacious, ergonomically designed cabin that allows for better patient access, making it easier for medical personnel to provide in-flight care.

- Range: location dependent, for rural areas longer range more desirable than in urban areas; short range also limiting as carrying dead fuel weight might not allow for sufficient payload on return (patient!)
- Payload: people (5x90 kg), fuel (minimum for 1 hour 15 min plus reserve), medical equipment (explore market needs, but at least 125 kg for fixed fittings not including seats and 125 for carry on equipment) – for a total of around 1200 kg
- Good performance in extreme conditions: high altitude, hot-and-high, adverse weather
- Safety: twin engine, built-in automation for reduced workload and automation. Built-in crash-resistant fuel tanks and energy-absorbing seats improve survivability in case of an accident. Designed with advanced obstacle detection and terrain awareness systems to improve safety when landing in difficult environments. Better integration of night vision systems and weather-resistant avionics to allow for safer operations in poor visibility conditions.
- Resilient to land in unprepared terrain, minimizing downwash and with considerations for potential debris, brownout and damage to nearby structures.
- Comfort: reduced vibrations and noise for enhanced patient and crew comfort and facilitate medical care.
- Low operational cost and ease of maintenance.

Project Outcomes

- *Vehicle Design Specification* report: A single collaborative document outlining the proposed concept as a whole. (5 pages max)
- Individual reports (12 pages max). Technical report detailing the student's contribution to the project
- FATT presentations: mid-project individual presentation and final group presentation.
- Project poster. To be presented in the last week of the project. Details to be announced.

Project management

Coordination

FATT01 members will act as coordinators to facilitate the running of the project. They will also act as mediators to solve design conflicts that may arise between FATTs, but if an external decision is needed, it will need to be taken by the whole team through an appropriate voting mechanism.

Each student will be assigned an academic advisor that will run weekly meetings with the team members.

Group meetings

Weekly group meetings will be set up with the participation of all project members and the academic advisors. The meetings will be called by FATT01, who will organize the agenda according to the issues that need discussion. A schedule to be available to the academic advisors at the start. All members of the team will present at least twice before the final presentation on 19/06/2025

Assessment

Assessment criteria follow the rules set by the Department.

Individual reports (50%)

- Each group member should produce an individual report. Report should be limited to 12 pages, plus appendices.
- Report should include as an executive summary the contribution of each the member to the collective Vehicle Design Specification Report.

Carrying-out (35%)

- Each FATT will prepare a presentation weekly group meeting. Each FATT member will be given 10 minutes to present, followed by a 5-minute Q&A. Not all members of a FATT need to present each week, however each member must have presented at least twice by the end of the project
- 10% of the carrying-out mark will be decided by the team members through a peer assessment exercise in the last week of the project.

Final presentation (15%)

- A 1.5 hour group presentation, detailing the GDP group's solution to the given design brief. A 30 minute Q&A session will follow.
- The presentations are marked as follows: Clarity 20%, Visual Aids 20%, Explanation of Problems 30%, Technical Challenge and Solution 30%.

Functional-Area Technical Teams

FATT01 – Project coordinators

Academic advisor: Dr Ribera Vicent

- Project coordination & information management
- Cost estimation and market analysis
- Conceptual trade off studies & design
- Compliance with all relevant design standards and regulations

FATT02 – Aerodynamics

Academic advisor: Dr Steiros

- Rotor(s) design
- Design of lifting surfaces
- Airfoil selection/design/optimizations
- Fuselage and empennage aerodynamic design.
- If present, tail rotor design.

FATT03 – Structural Design & CAD

Academic advisor: Dr Shamsuddin

- Prediction of design loading conditions
- Rotor structural design and dynamic blade behaviour
- Selection of aircraft structural layout and sizing of major components
- Material and manufacturing method selection
- Landing gear design
- Hub design
- CAD modeling

FATT04 –Systems & Propulsion

Academic advisor: Dr Fasel, Dr Seifikar

- Powerplant type selection & integration
- Powerplant sizing and performance prediction
- Transmission considerations
- Vehicle sub-systems selection/sizing, layout and integration
- Sensors and avionics
- Weight & Balance Prediction
- Medical and mission specific equipment

FATT05 – Flight Mechanics, Performance & Control

Academic advisor: Dr Armanini, Dr Ribera Vicent

- Mission definition and operations
- Vehicle static and dynamic stability prediction
- Response to control inputs
- Design of flight control system
- Aircraft performance prediction